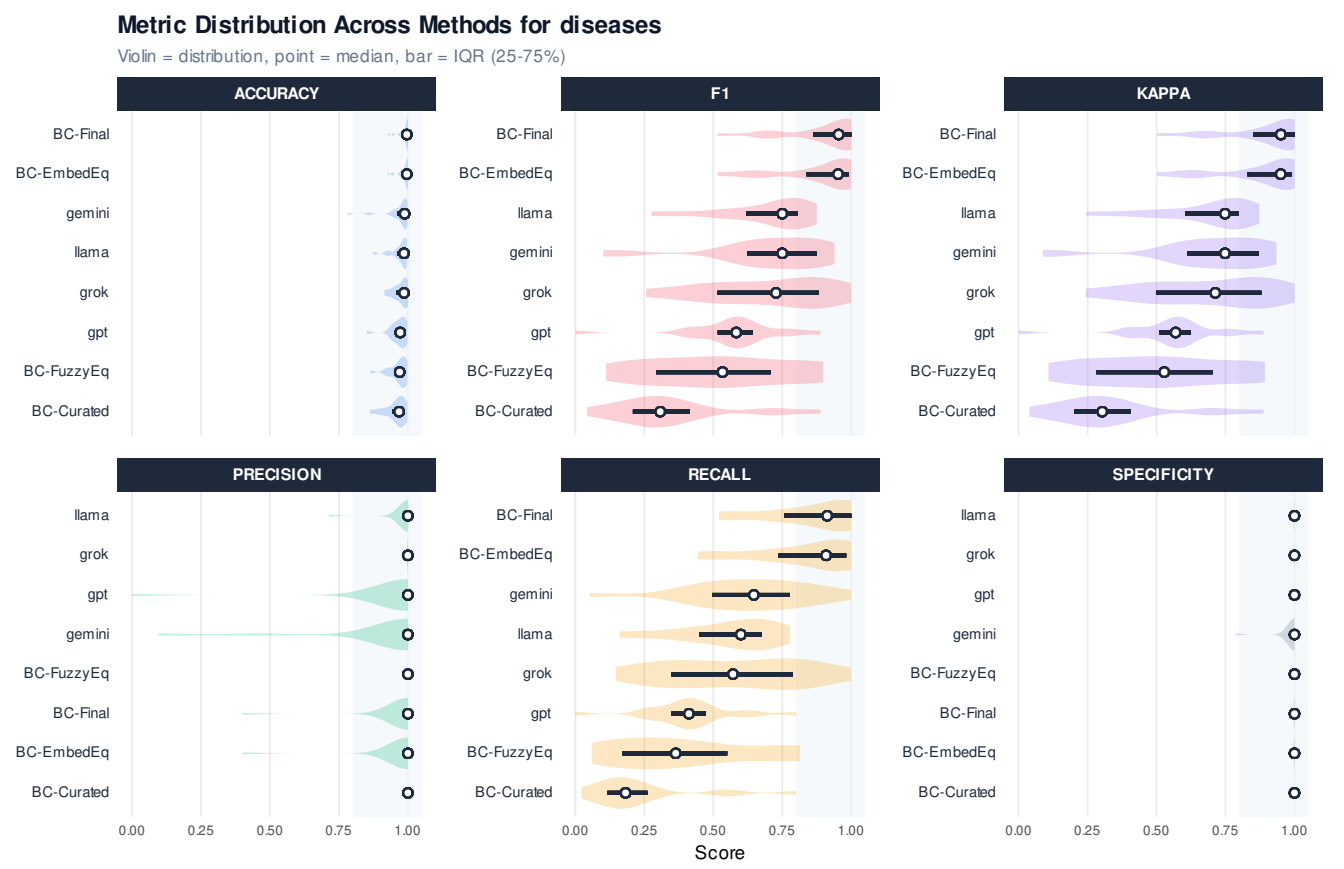
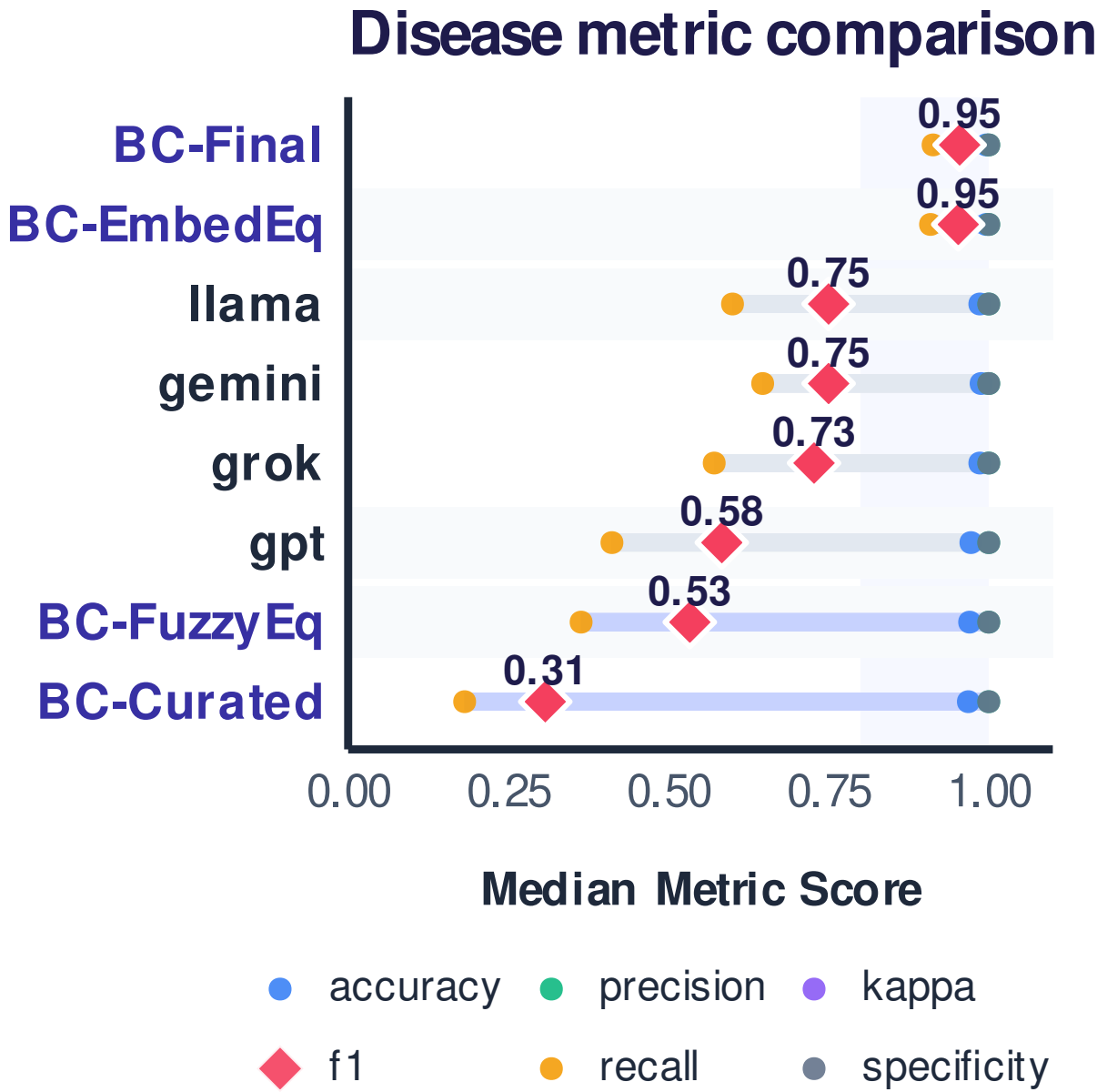

Metric Distribution Across Methods for diseases
Violin = distribution, point = median, bar = IQR (25-75%)
ACCURACY
F1
KAPPA
PRECISION
RECALL
SPECIFICITY
BC-Final
BC-EmbedEq
gemini
llama
grok
gpt
BC-FuzzyEq
BC-Curated
0.00
0.25
0.50
0.75
1.00
Score


Disease metric comparison
BC-Final
BC-EmbedEq
llama
gemini
grok
gpt
BC-FuzzyEq
BC-Curated
0.95
0.95
0.75
0.75
0.73
0.58
0.53
0.31
0.00
0.25
0.50
0.75
1.00
Median Metric Score
accuracy
precision
kappa
f1
recall
specificity

# Disease Synonyms Extraction: Comparative Performance Matrix

Methods ranked by Median F1

Disease

### BC-Curated
**Rank 8 | Med F1: 0.31 | R: 0.18 | P: 1.00 | K: 0.30**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.02 | 0.04 | 0.91 | 1.00 | 0.04 |
| GESTATIONAL DIABETES | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| ASTHMA | 1.00 | 0.30 | 0.46 | 0.96 | 1.00 | 0.44 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.17 | 0.29 | 0.97 | 1.00 | 0.28 |
| GASTRIC ULCER | 1.00 | 0.54 | 0.70 | 0.97 | 1.00 | 0.69 |
| RUBELLA | 1.00 | 0.11 | 0.20 | 0.97 | 1.00 | 0.19 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 0.20 | 0.33 | 0.99 | 1.00 | 0.33 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 0.20 | 0.33 | 0.99 | 1.00 | 0.33 |
| DIPHTHERIA | 1.00 | 0.18 | 0.31 | 0.98 | 1.00 | 0.30 |
| COVID-19 | 1.00 | 0.12 | 0.21 | 0.97 | 1.00 | 0.20 |
| OSTEOARTHRITIS | 1.00 | 0.17 | 0.29 | 0.94 | 1.00 | 0.28 |
| HORDEOLUM | 1.00 | 0.25 | 0.40 | 0.97 | 1.00 | 0.39 |
| NEONATAL JAUNDICE | 1.00 | 0.56 | 0.71 | 0.99 | 1.00 | 0.71 |
| MULTIPLE SCLEROSIS | 1.00 | 0.15 | 0.26 | 0.92 | 1.00 | 0.24 |
| PSORIASIS | 1.00 | 0.05 | 0.10 | 0.89 | 1.00 | 0.09 |
| MULTIPLE MYELOMA | 1.00 | 0.12 | 0.21 | 0.95 | 1.00 | 0.20 |
| FEVER | 1.00 | 0.27 | 0.43 | 0.98 | 1.00 | 0.42 |
| DOWN SYNDROME | 1.00 | 0.21 | 0.34 | 0.95 | 1.00 | 0.33 |
| HYPERTHYROIDISM | 1.00 | 0.08 | 0.15 | 0.86 | 1.00 | 0.13 |

### BC-FuzzyEq
**Rank 7 | Med F1: 0.53 | R: 0.36 | P: 1.00 | K: 0.53**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.15 | 0.26 | 0.92 | 1.00 | 0.25 |
| GESTATIONAL DIABETES | 1.00 | 0.60 | 0.75 | 1.00 | 1.00 | 0.75 |
| ASTHMA | 1.00 | 0.81 | 0.90 | 0.99 | 1.00 | 0.89 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.17 | 0.29 | 0.97 | 1.00 | 0.28 |
| GASTRIC ULCER | 1.00 | 0.80 | 0.89 | 0.99 | 1.00 | 0.88 |
| RUBELLA | 1.00 | 0.50 | 0.67 | 0.98 | 1.00 | 0.66 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| DIPHTHERIA | 1.00 | 0.36 | 0.53 | 0.99 | 1.00 | 0.53 |
| COVID-19 | 1.00 | 0.06 | 0.11 | 0.97 | 1.00 | 0.11 |
| OSTEOARTHRITIS | 1.00 | 0.49 | 0.65 | 0.96 | 1.00 | 0.64 |
| HORDEOLUM | 1.00 | 0.25 | 0.40 | 0.97 | 1.00 | 0.39 |
| NEONATAL JAUNDICE | 1.00 | 0.11 | 0.20 | 0.98 | 1.00 | 0.20 |
| MULTIPLE SCLEROSIS | 1.00 | 0.38 | 0.55 | 0.94 | 1.00 | 0.53 |
| PSORIASIS | 1.00 | 0.31 | 0.47 | 0.92 | 1.00 | 0.44 |
| MULTIPLE MYELOMA | 1.00 | 0.35 | 0.51 | 0.97 | 1.00 | 0.50 |
| FEVER | 1.00 | 0.36 | 0.53 | 0.99 | 1.00 | 0.53 |
| DOWN SYNDROME | 1.00 | 0.18 | 0.30 | 0.94 | 1.00 | 0.29 |
| HYPERTHYROIDISM | 1.00 | 0.09 | 0.17 | 0.86 | 1.00 | 0.15 |

### gpt-4o-mini
**Rank 6 | Med F1: 0.58 | R: 0.41 | P: 1.00 | K: 0.57**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.41 | 0.58 | 0.95 | 1.00 | 0.56 |
| GESTATIONAL DIABETES | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| ASTHMA | 1.00 | 0.48 | 0.65 | 0.97 | 1.00 | 0.64 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.61 | 0.76 | 0.99 | 1.00 | 0.75 |
| GASTRIC ULCER | 1.00 | 0.29 | 0.44 | 0.95 | 1.00 | 0.43 |
| RUBELLA | 1.00 | 0.33 | 0.50 | 0.98 | 1.00 | 0.49 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 0.40 | 0.57 | 0.99 | 1.00 | 0.57 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 0.40 | 0.57 | 0.99 | 1.00 | 0.57 |
| DIPHTHERIA | 1.00 | 0.64 | 0.78 | 0.99 | 1.00 | 0.77 |
| COVID-19 | 1.00 | 0.41 | 0.58 | 0.98 | 1.00 | 0.57 |
| OSTEOARTHRITIS | 1.00 | 0.49 | 0.65 | 0.96 | 1.00 | 0.64 |
| HORDEOLUM | 1.00 | 0.25 | 0.40 | 0.97 | 1.00 | 0.39 |
| NEONATAL JAUNDICE | 1.00 | 0.44 | 0.62 | 0.99 | 1.00 | 0.61 |
| MULTIPLE SCLEROSIS | 1.00 | 0.38 | 0.55 | 0.94 | 1.00 | 0.53 |
| PSORIASIS | 1.00 | 0.47 | 0.64 | 0.94 | 1.00 | 0.61 |
| MULTIPLE MYELOMA | 1.00 | 0.42 | 0.59 | 0.97 | 1.00 | 0.58 |
| FEVER | 1.00 | 0.36 | 0.53 | 0.99 | 1.00 | 0.53 |
| DOWN SYNDROME | 1.00 | 0.26 | 0.42 | 0.95 | 1.00 | 0.40 |
| HYPERTHYROIDISM | 0.00 | 0.00 | 0.00 | 0.85 | 1.00 | 0.00 |

### grok-4.1-fast-non-reasoning
**Rank 5 | Med F1: 0.73 | R: 0.57 | P: 1.00 | K: 0.71**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.85 | 0.92 | 0.99 | 1.00 | 0.91 |
| GESTATIONAL DIABETES | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ASTHMA | 1.00 | 0.74 | 0.85 | 0.99 | 1.00 | 0.84 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.78 | 0.88 | 0.99 | 1.00 | 0.87 |
| GASTRIC ULCER | 1.00 | 0.31 | 0.48 | 0.95 | 1.00 | 0.46 |
| RUBELLA | 1.00 | 0.89 | 0.94 | 1.00 | 1.00 | 0.94 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| DIPHTHERIA | 1.00 | 0.73 | 0.84 | 0.99 | 1.00 | 0.84 |
| COVID-19 | 1.00 | 0.65 | 0.79 | 0.99 | 1.00 | 0.78 |
| OSTEOARTHRITIS | 1.00 | 0.57 | 0.73 | 0.97 | 1.00 | 0.71 |
| HORDEOLUM | 1.00 | 0.30 | 0.46 | 0.97 | 1.00 | 0.45 |
| NEONATAL JAUNDICE | 1.00 | 0.33 | 0.50 | 0.99 | 1.00 | 0.50 |
| MULTIPLE SCLEROSIS | 1.00 | 0.53 | 0.69 | 0.96 | 1.00 | 0.67 |
| PSORIASIS | 1.00 | 0.36 | 0.53 | 0.93 | 1.00 | 0.50 |
| MULTIPLE MYELOMA | 1.00 | 0.19 | 0.32 | 0.96 | 1.00 | 0.31 |
| FEVER | 1.00 | 0.45 | 0.62 | 0.99 | 1.00 | 0.62 |
| DOWN SYNDROME | 1.00 | 0.15 | 0.26 | 0.94 | 1.00 | 0.24 |
| HYPERTHYROIDISM | 0.97 | 0.44 | 0.61 | 0.91 | 1.00 | 0.56 |

### gemini-2.5-flash-lite
**Rank 3 | Med F1: 0.75 | R: 0.65 | P: 1.00 | K: 0.75**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 0.98 | 0.89 | 0.93 | 0.99 | 1.00 | 0.93 |
| GESTATIONAL DIABETES | 1.00 | 0.80 | 0.89 | 1.00 | 1.00 | 0.89 |
| ASTHMA | 1.00 | 0.78 | 0.88 | 0.99 | 1.00 | 0.87 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.78 | 0.88 | 0.99 | 1.00 | 0.87 |
| GASTRIC ULCER | 1.00 | 0.89 | 0.94 | 0.99 | 1.00 | 0.94 |
| RUBELLA | 1.00 | 0.78 | 0.88 | 0.99 | 1.00 | 0.87 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 0.40 | 0.57 | 0.99 | 1.00 | 0.57 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 0.60 | 0.75 | 1.00 | 1.00 | 0.75 |
| DIPHTHERIA | 1.00 | 0.55 | 0.71 | 0.99 | 1.00 | 0.70 |
| COVID-19 | 0.10 | 0.65 | 0.17 | 0.78 | 0.79 | 0.11 |
| OSTEOARTHRITIS | 1.00 | 0.46 | 0.63 | 0.96 | 1.00 | 0.61 |
| HORDEOLUM | 1.00 | 0.65 | 0.79 | 0.99 | 1.00 | 0.78 |
| NEONATAL JAUNDICE | 0.45 | 1.00 | 0.62 | 0.98 | 0.98 | 0.61 |
| MULTIPLE SCLEROSIS | 1.00 | 0.51 | 0.68 | 0.95 | 1.00 | 0.65 |
| PSORIASIS | 1.00 | 0.48 | 0.65 | 0.94 | 1.00 | 0.62 |
| MULTIPLE MYELOMA | 1.00 | 0.65 | 0.79 | 0.98 | 1.00 | 0.78 |
| FEVER | 1.00 | 0.64 | 0.78 | 0.99 | 1.00 | 0.77 |
| DOWN SYNDROME | 1.00 | 0.41 | 0.58 | 0.96 | 1.00 | 0.57 |
| HYPERTHYROIDISM | 1.00 | 0.05 | 0.10 | 0.86 | 1.00 | 0.09 |

### llama-3.3-70b-versatile
**Rank 4 | Med F1: 0.75 | R: 0.60 | P: 1.00 | K: 0.75**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.76 | 0.86 | 0.98 | 1.00 | 0.85 |
| GESTATIONAL DIABETES | 1.00 | 0.60 | 0.75 | 1.00 | 1.00 | 0.75 |
| ASTHMA | 1.00 | 0.67 | 0.80 | 0.98 | 1.00 | 0.79 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.72 | 0.84 | 0.99 | 1.00 | 0.83 |
| GASTRIC ULCER | 1.00 | 0.69 | 0.81 | 0.98 | 1.00 | 0.80 |
| RUBELLA | 1.00 | 0.67 | 0.80 | 0.99 | 1.00 | 0.79 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 0.60 | 0.75 | 1.00 | 1.00 | 0.75 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 0.60 | 0.75 | 1.00 | 1.00 | 0.75 |
| DIPHTHERIA | 1.00 | 0.55 | 0.71 | 0.99 | 1.00 | 0.70 |
| COVID-19 | 1.00 | 0.71 | 0.83 | 0.99 | 1.00 | 0.82 |
| OSTEOARTHRITIS | 1.00 | 0.43 | 0.60 | 0.96 | 1.00 | 0.58 |
| HORDEOLUM | 1.00 | 0.65 | 0.79 | 0.99 | 1.00 | 0.78 |
| NEONATAL JAUNDICE | 1.00 | 0.78 | 0.88 | 1.00 | 1.00 | 0.87 |
| MULTIPLE SCLEROSIS | 0.71 | 0.43 | 0.53 | 0.93 | 0.98 | 0.50 |
| PSORIASIS | 1.00 | 0.34 | 0.51 | 0.92 | 1.00 | 0.48 |
| MULTIPLE MYELOMA | 1.00 | 0.23 | 0.38 | 0.96 | 1.00 | 0.36 |
| FEVER | 1.00 | 0.55 | 0.71 | 0.99 | 1.00 | 0.70 |
| DOWN SYNDROME | 1.00 | 0.47 | 0.64 | 0.96 | 1.00 | 0.62 |
| HYPERTHYROIDISM | 1.00 | 0.16 | 0.28 | 0.87 | 1.00 | 0.24 |

### BC-EmbedEq
**Rank 2 | Med F1: 0.95 | R: 0.91 | P: 1.00 | K: 0.95**

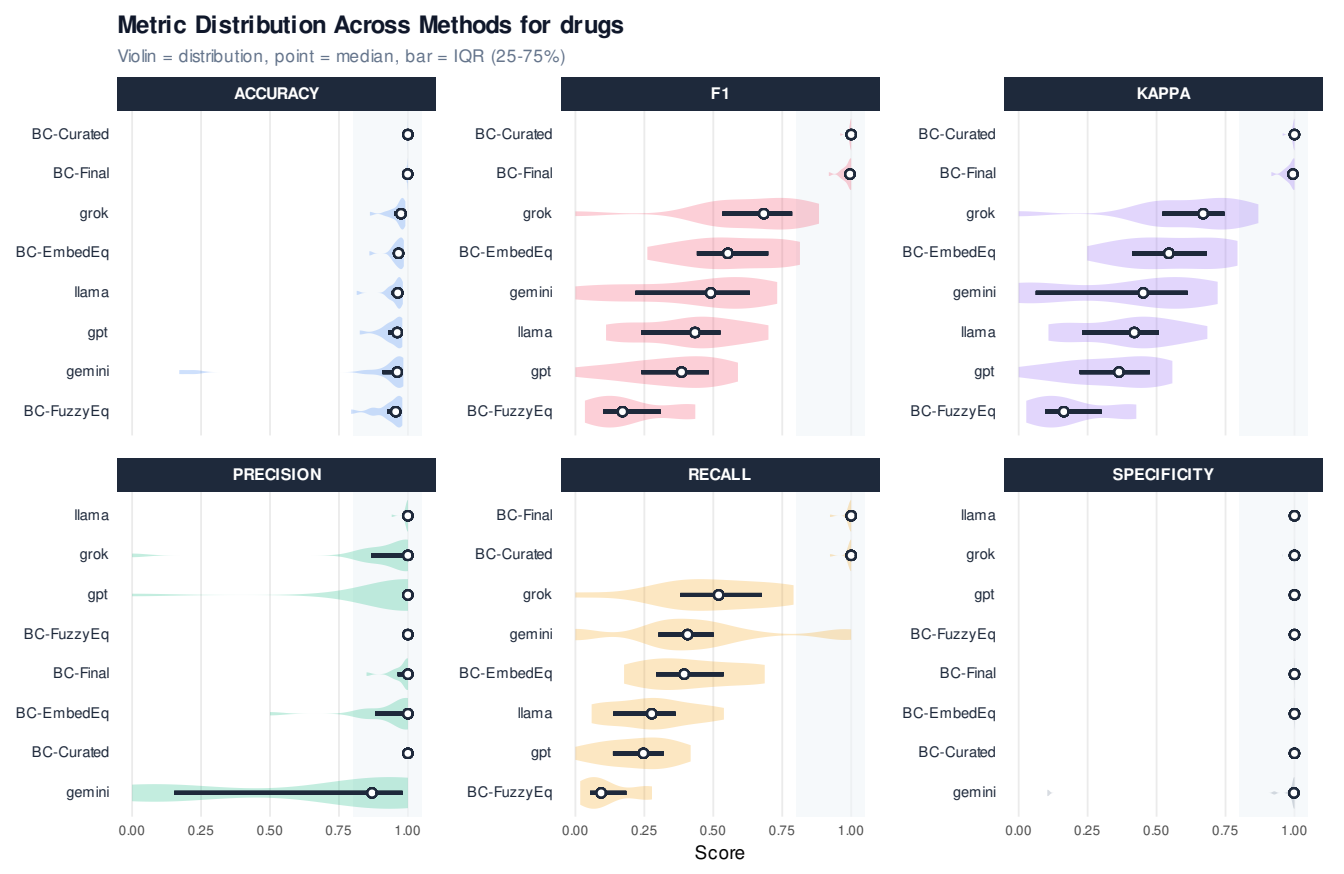
| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.91 | 0.95 | 0.99 | 1.00 | 0.95 |
| GESTATIONAL DIABETES | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ASTHMA | 1.00 | 0.96 | 0.98 | 1.00 | 1.00 | 0.98 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.89 | 0.94 | 1.00 | 1.00 | 0.94 |
| GASTRIC ULCER | 1.00 | 0.94 | 0.97 | 1.00 | 1.00 | 0.97 |
| RUBELLA | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| DIPHTHERIA | 1.00 | 0.91 | 0.95 | 1.00 | 1.00 | 0.95 |
| COVID-19 | 1.00 | 0.94 | 0.97 | 1.00 | 1.00 | 0.97 |
| OSTEOARTHRITIS | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| HORDEOLUM | 1.00 | 0.70 | 0.82 | 0.99 | 1.00 | 0.82 |
| NEONATAL JAUNDICE | 1.00 | 0.44 | 0.62 | 0.99 | 1.00 | 0.61 |
| MULTIPLE SCLEROSIS | 1.00 | 0.74 | 0.85 | 0.98 | 1.00 | 0.84 |
| PSORIASIS | 1.00 | 0.52 | 0.68 | 0.94 | 1.00 | 0.65 |
| MULTIPLE MYELOMA | 1.00 | 0.77 | 0.87 | 0.99 | 1.00 | 0.86 |
| FEVER | 0.40 | 0.73 | 0.52 | 0.97 | 0.98 | 0.50 |
| DOWN SYNDROME | 1.00 | 0.88 | 0.94 | 0.99 | 1.00 | 0.93 |
| HYPERTHYROIDISM | 1.00 | 0.51 | 0.67 | 0.93 | 1.00 | 0.64 |

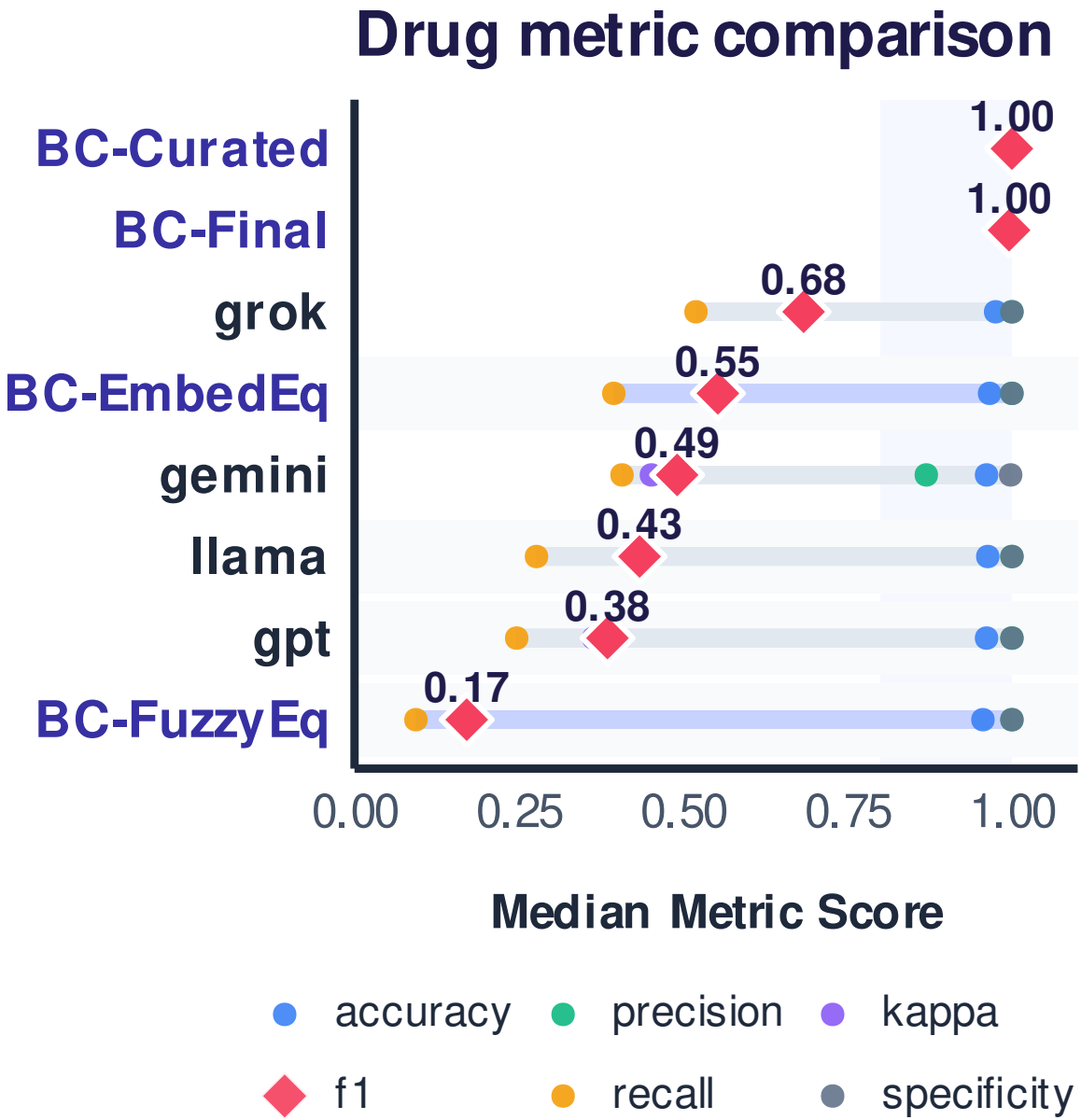
### BC-Final
**Rank 1 | Med F1: 0.95 | R: 0.91 | P: 1.00 | K: 0.95**

| Disease | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| UROLITHIASIS | 1.00 | 0.91 | 0.95 | 0.99 | 1.00 | 0.95 |
| GESTATIONAL DIABETES | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ASTHMA | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ESSENTIAL HYPERTENSION | 1.00 | 0.89 | 0.94 | 1.00 | 1.00 | 0.94 |
| GASTRIC ULCER | 1.00 | 0.94 | 0.97 | 1.00 | 1.00 | 0.97 |
| RUBELLA | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| IRON DEFICIENCY ANAEMIA | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| GENERALIZED ANXIETY DISORDER | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| DIPHTHERIA | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| COVID-19 | 1.00 | 0.94 | 0.97 | 1.00 | 1.00 | 0.97 |
| OSTEOARTHRITIS | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| HORDEOLUM | 1.00 | 0.80 | 0.89 | 0.99 | 1.00 | 0.88 |
| NEONATAL JAUNDICE | 1.00 | 0.56 | 0.71 | 0.99 | 1.00 | 0.71 |
| MULTIPLE SCLEROSIS | 1.00 | 0.74 | 0.85 | 0.98 | 1.00 | 0.84 |
| PSORIASIS | 1.00 | 0.53 | 0.70 | 0.95 | 1.00 | 0.67 |
| MULTIPLE MYELOMA | 1.00 | 0.77 | 0.87 | 0.99 | 1.00 | 0.86 |
| FEVER | 0.40 | 0.73 | 0.52 | 0.97 | 0.98 | 0.50 |
| DOWN SYNDROME | 1.00 | 0.91 | 0.95 | 0.99 | 1.00 | 0.95 |
| HYPERTHYROIDISM | 1.00 | 0.52 | 0.68 | 0.93 | 1.00 | 0.65 |

Performance Score

0.00 0.25 0.50 0.75 1.00


Metric Distribution Across Methods for drugs
Violin = distribution, point = median, bar = IQR (25-75%)
ACCURACY
F1
KAPPA
PRECISION
RECALL
SPECIFICITY
BC-Curated
BC-Final
grok
BC-EmbedEq
llama
gpt
gemini
BC-FuzzyEq
0.00
0.25
0.50
0.75
1.00
Score


Drug metric comparison
BC-Curated
BC-Final
grok
BC-EmbedEq
gemini
llama
gpt
BC-FuzzyEq
1.00
1.00
0.68
0.55
0.49
0.43
0.38
0.17
0.00
0.25
0.50
0.75
1.00
Median Metric Score
accuracy
precision
kappa
f1
recall
specificity

# Drug Synonyms Extraction: Comparative Performance Matrix

Methods ranked by Median F1

Drug

### BC-FuzzyEq
Rank 8 | Med F1: 0.17 | R: 0.09 | P: 1.00 | K: 0.16

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 0.03 | 0.06 | 0.88 | 1.00 | 0.05 |
| DENOSUMAB | 1.00 | 0.21 | 0.34 | 0.97 | 1.00 | 0.33 |
| METHOTREXATE | 1.00 | 0.10 | 0.19 | 0.95 | 1.00 | 0.18 |
| OMEPRAZOLE | 1.00 | 0.11 | 0.20 | 0.97 | 1.00 | 0.19 |
| SIMVASTATIN | 1.00 | 0.28 | 0.43 | 0.98 | 1.00 | 0.43 |
| IVERMECTIN | 1.00 | 0.11 | 0.20 | 0.92 | 1.00 | 0.18 |
| SALBUTAMOL | 1.00 | 0.05 | 0.10 | 0.93 | 1.00 | 0.09 |
| LATANOPROST | 1.00 | 0.27 | 0.42 | 0.98 | 1.00 | 0.41 |
| CIPROFLOXACIN | 1.00 | 0.27 | 0.42 | 0.98 | 1.00 | 0.41 |
| HYDROCORTISONE | 1.00 | 0.02 | 0.03 | 0.79 | 1.00 | 0.03 |
| ASPIRIN | 1.00 | 0.03 | 0.05 | 0.87 | 1.00 | 0.05 |
| DIAZEPAM | 1.00 | 0.07 | 0.13 | 0.93 | 1.00 | 0.12 |
| PREDNISOLONE | 1.00 | 0.08 | 0.15 | 0.96 | 1.00 | 0.15 |
| IOHEXOL | 1.00 | 0.06 | 0.11 | 0.97 | 1.00 | 0.11 |

### gpt-4o-mini
Rank 7 | Med F1: 0.38 | R: 0.25 | P: 1.00 | K: 0.36

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 0.42 | 0.59 | 0.93 | 1.00 | 0.56 |
| DENOSUMAB | 1.00 | 0.38 | 0.55 | 0.97 | 1.00 | 0.53 |
| METHOTREXATE | 1.00 | 0.34 | 0.51 | 0.97 | 1.00 | 0.50 |
| OMEPRAZOLE | 1.00 | 0.28 | 0.43 | 0.98 | 1.00 | 0.43 |
| SIMVASTATIN | 1.00 | 0.33 | 0.50 | 0.98 | 1.00 | 0.49 |
| IVERMECTIN | 1.00 | 0.13 | 0.23 | 0.93 | 1.00 | 0.22 |
| SALBUTAMOL | 1.00 | 0.15 | 0.27 | 0.94 | 1.00 | 0.25 |
| LATANOPROST | 1.00 | 0.27 | 0.42 | 0.98 | 1.00 | 0.41 |
| CIPROFLOXACIN | 0.80 | 0.27 | 0.40 | 0.98 | 1.00 | 0.39 |
| HYDROCORTISONE | 1.00 | 0.17 | 0.29 | 0.83 | 1.00 | 0.24 |
| ASPIRIN | 1.00 | 0.23 | 0.37 | 0.89 | 1.00 | 0.34 |
| DIAZEPAM | 1.00 | 0.09 | 0.17 | 0.93 | 1.00 | 0.16 |
| PREDNISOLONE | 1.00 | 0.04 | 0.08 | 0.96 | 1.00 | 0.08 |
| IOHEXOL | 0.00 | 0.00 | 0.00 | 0.97 | 1.00 | 0.00 |

### llama-3.3-70b-versatile
Rank 6 | Med F1: 0.43 | R: 0.28 | P: 1.00 | K: 0.42

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 0.46 | 0.63 | 0.93 | 1.00 | 0.60 |
| DENOSUMAB | 1.00 | 0.21 | 0.34 | 0.97 | 1.00 | 0.33 |
| METHOTREXATE | 1.00 | 0.28 | 0.43 | 0.96 | 1.00 | 0.42 |
| OMEPRAZOLE | 1.00 | 0.11 | 0.20 | 0.97 | 1.00 | 0.19 |
| SIMVASTATIN | 1.00 | 0.28 | 0.43 | 0.98 | 1.00 | 0.43 |
| IVERMECTIN | 1.00 | 0.28 | 0.44 | 0.94 | 1.00 | 0.42 |
| SALBUTAMOL | 1.00 | 0.54 | 0.70 | 0.97 | 1.00 | 0.68 |
| LATANOPROST | 1.00 | 0.27 | 0.42 | 0.98 | 1.00 | 0.41 |
| CIPROFLOXACIN | 1.00 | 0.33 | 0.50 | 0.98 | 1.00 | 0.49 |
| HYDROCORTISONE | 1.00 | 0.11 | 0.20 | 0.81 | 1.00 | 0.17 |
| ASPIRIN | 1.00 | 0.45 | 0.62 | 0.92 | 1.00 | 0.59 |
| DIAZEPAM | 0.94 | 0.37 | 0.53 | 0.95 | 1.00 | 0.51 |
| PREDNISOLONE | 1.00 | 0.08 | 0.15 | 0.96 | 1.00 | 0.15 |
| IOHEXOL | 1.00 | 0.06 | 0.11 | 0.97 | 1.00 | 0.11 |

### gemini-2.5-flash-lite
Rank 5 | Med F1: 0.49 | R: 0.41 | P: 0.87 | K: 0.45

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 0.49 | 0.66 | 0.94 | 1.00 | 0.63 |
| DENOSUMAB | 0.88 | 0.62 | 0.73 | 0.98 | 1.00 | 0.72 |
| METHOTREXATE | 1.00 | 0.41 | 0.59 | 0.97 | 1.00 | 0.57 |
| OMEPRAZOLE | 1.00 | 0.50 | 0.67 | 0.98 | 1.00 | 0.66 |
| SIMVASTATIN | 1.00 | 0.33 | 0.50 | 0.98 | 1.00 | 0.49 |
| IVERMECTIN | 0.92 | 0.50 | 0.65 | 0.95 | 1.00 | 0.63 |
| SALBUTAMOL | 0.92 | 0.28 | 0.43 | 0.95 | 1.00 | 0.41 |
| LATANOPROST | 0.86 | 0.40 | 0.55 | 0.98 | 1.00 | 0.54 |
| CIPROFLOXACIN | 0.00 | 0.00 | 0.00 | 0.97 | 1.00 | 0.00 |
| HYDROCORTISONE | 0.81 | 0.34 | 0.48 | 0.85 | 0.98 | 0.41 |
| ASPIRIN | 0.15 | 1.00 | 0.26 | 0.23 | 0.11 | 0.03 |
| DIAZEPAM | 0.08 | 0.95 | 0.15 | 0.17 | 0.10 | 0.01 |
| PREDNISOLONE | 0.16 | 0.29 | 0.20 | 0.90 | 0.93 | 0.15 |
| IOHEXOL | 0.00 | 0.00 | 0.00 | 0.97 | 1.00 | 0.00 |

### BC-EmbedEq
Rank 4 | Med F1: 0.55 | R: 0.39 | P: 1.00 | K: 0.54

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 0.69 | 0.81 | 0.96 | 1.00 | 0.79 |
| DENOSUMAB | 1.00 | 0.67 | 0.80 | 0.99 | 1.00 | 0.79 |
| METHOTREXATE | 1.00 | 0.52 | 0.68 | 0.97 | 1.00 | 0.67 |
| OMEPRAZOLE | 1.00 | 0.39 | 0.56 | 0.98 | 1.00 | 0.55 |
| SIMVASTATIN | 1.00 | 0.50 | 0.67 | 0.98 | 1.00 | 0.66 |
| IVERMECTIN | 1.00 | 0.54 | 0.70 | 0.96 | 1.00 | 0.69 |
| SALBUTAMOL | 1.00 | 0.26 | 0.41 | 0.95 | 1.00 | 0.39 |
| LATANOPROST | 1.00 | 0.33 | 0.50 | 0.98 | 1.00 | 0.49 |
| CIPROFLOXACIN | 0.86 | 0.40 | 0.55 | 0.98 | 1.00 | 0.54 |
| HYDROCORTISONE | 0.95 | 0.35 | 0.51 | 0.86 | 1.00 | 0.45 |
| ASPIRIN | 0.96 | 0.68 | 0.80 | 0.95 | 1.00 | 0.77 |
| DIAZEPAM | 0.86 | 0.28 | 0.42 | 0.94 | 1.00 | 0.40 |
| PREDNISOLONE | 0.83 | 0.21 | 0.33 | 0.96 | 1.00 | 0.32 |
| IOHEXOL | 0.50 | 0.18 | 0.26 | 0.97 | 0.99 | 0.25 |

### grok-4.1-fast-non-reasoning
Rank 3 | Med F1: 0.68 | R: 0.52 | P: 1.00 | K: 0.67

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 0.79 | 0.88 | 0.97 | 1.00 | 0.87 |
| DENOSUMAB | 1.00 | 0.58 | 0.74 | 0.98 | 1.00 | 0.73 |
| METHOTREXATE | 1.00 | 0.55 | 0.71 | 0.98 | 1.00 | 0.70 |
| OMEPRAZOLE | 0.93 | 0.72 | 0.81 | 0.99 | 1.00 | 0.81 |
| SIMVASTATIN | 1.00 | 0.56 | 0.71 | 0.99 | 1.00 | 0.71 |
| IVERMECTIN | 1.00 | 0.30 | 0.47 | 0.94 | 1.00 | 0.44 |
| SALBUTAMOL | 1.00 | 0.49 | 0.66 | 0.96 | 1.00 | 0.64 |
| LATANOPROST | 0.86 | 0.40 | 0.55 | 0.98 | 1.00 | 0.54 |
| CIPROFLOXACIN | 0.86 | 0.40 | 0.55 | 0.98 | 1.00 | 0.54 |
| HYDROCORTISONE | 0.82 | 0.78 | 0.80 | 0.92 | 0.96 | 0.75 |
| ASPIRIN | 0.00 | 0.00 | 0.00 | 0.86 | 1.00 | 0.00 |
| DIAZEPAM | 1.00 | 0.33 | 0.49 | 0.95 | 1.00 | 0.47 |
| PREDNISOLONE | 0.90 | 0.38 | 0.53 | 0.97 | 1.00 | 0.52 |
| IOHEXOL | 1.00 | 0.71 | 0.83 | 0.99 | 1.00 | 0.82 |

### BC-Final
Rank 2 | Med F1: 1.00 | R: 1.00 | P: 1.00 | K: 0.99

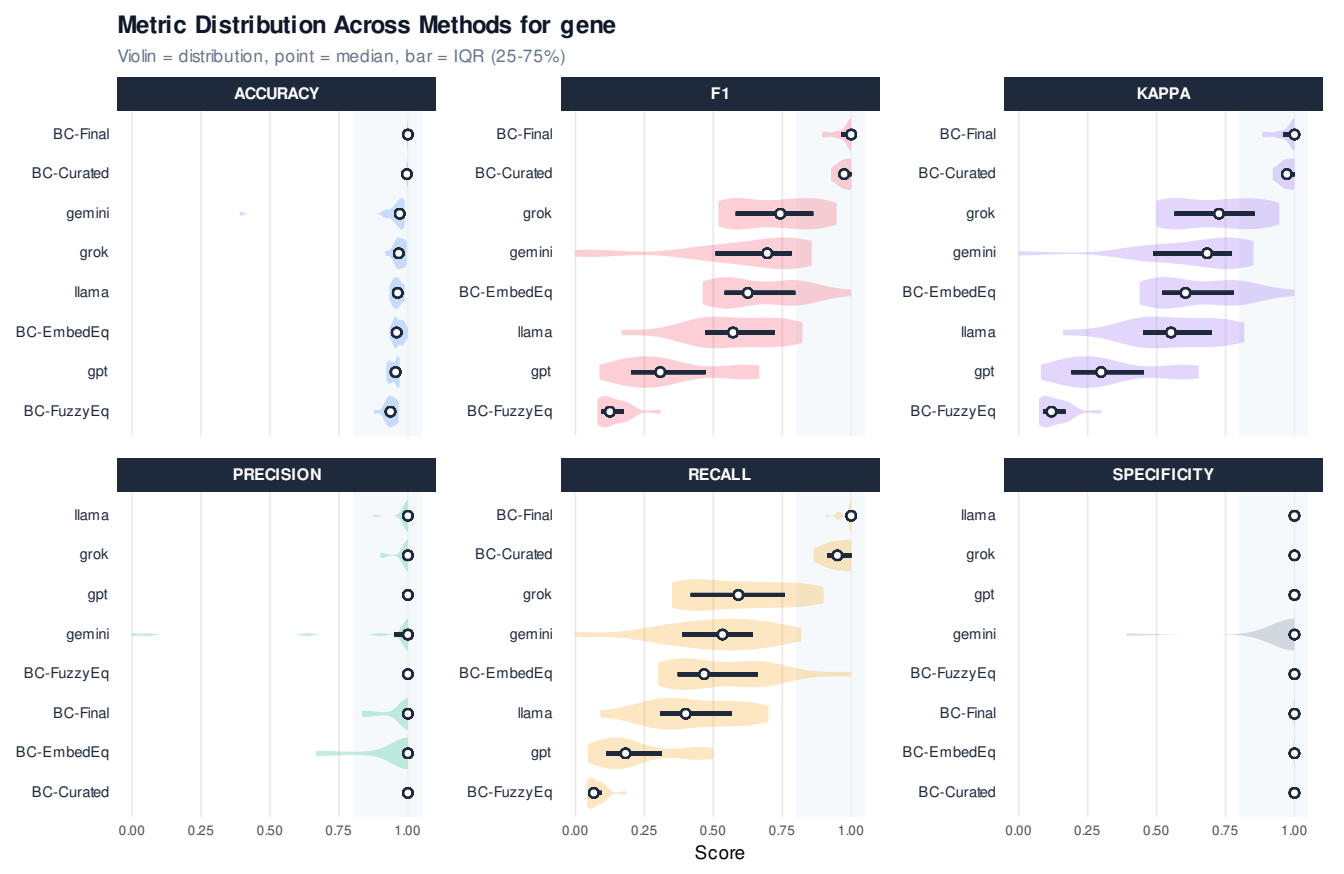
| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| DENOSUMAB | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| METHOTREXATE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| OMEPRAZOLE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| SIMVASTATIN | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| IVERMECTIN | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| SALBUTAMOL | 1.00 | 0.92 | 0.96 | 0.99 | 1.00 | 0.96 |
| LATANOPROST | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CIPROFLOXACIN | 0.94 | 1.00 | 0.97 | 1.00 | 1.00 | 0.97 |
| HYDROCORTISONE | 0.98 | 1.00 | 0.99 | 1.00 | 1.00 | 0.99 |
| ASPIRIN | 0.97 | 1.00 | 0.99 | 1.00 | 1.00 | 0.98 |
| DIAZEPAM | 0.96 | 1.00 | 0.98 | 1.00 | 1.00 | 0.98 |
| PREDNISOLONE | 0.96 | 1.00 | 0.98 | 1.00 | 1.00 | 0.98 |
| IOHEXOL | 0.85 | 1.00 | 0.92 | 0.99 | 0.99 | 0.92 |

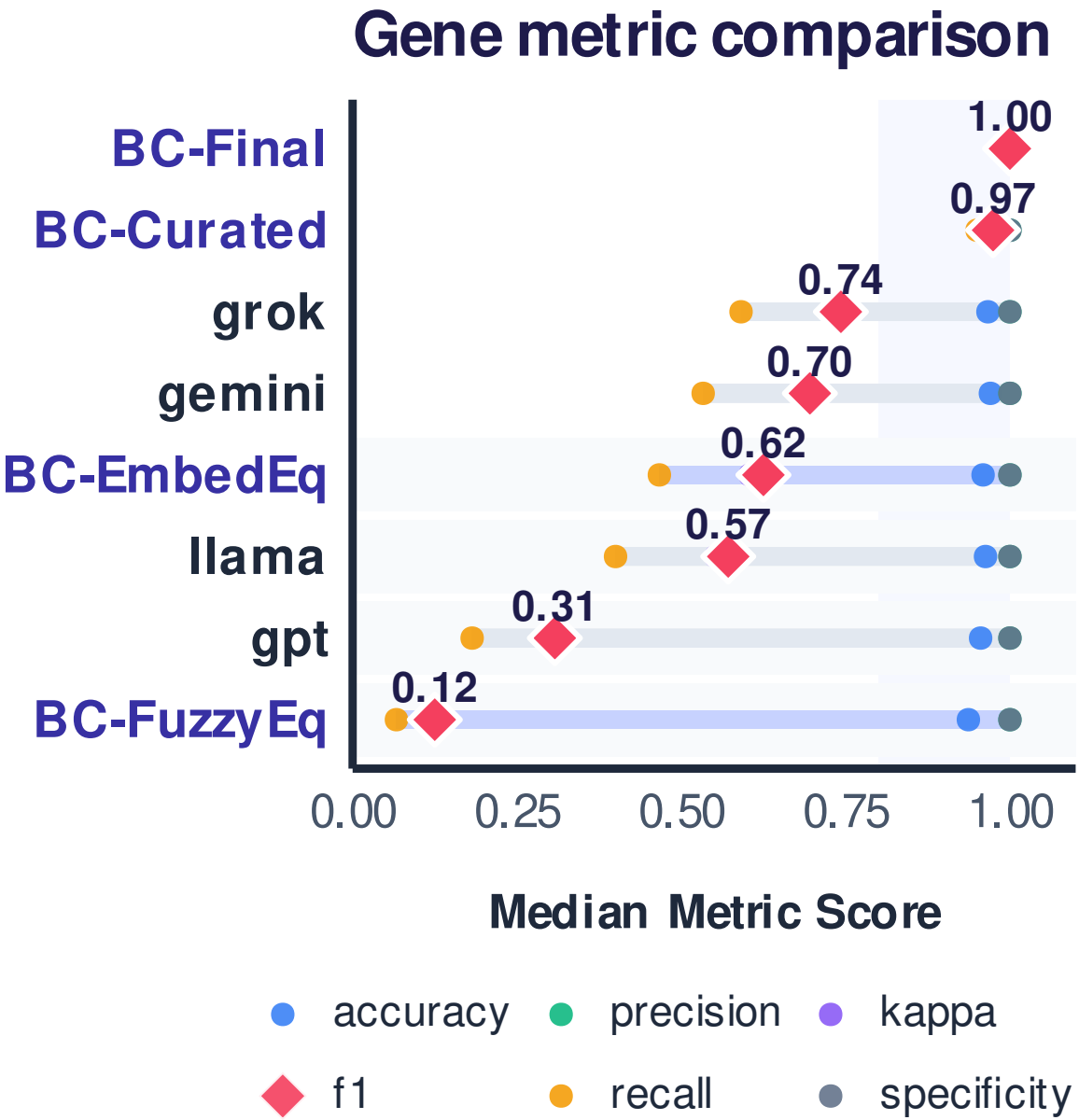
### BC-Curated
Rank 1 | Med F1: 1.00 | R: 1.00 | P: 1.00 | K: 1.00

| Drug | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| LEVONORGESTREL | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| DENOSUMAB | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| METHOTREXATE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| OMEPRAZOLE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| SIMVASTATIN | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| IVERMECTIN | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| SALBUTAMOL | 1.00 | 0.92 | 0.96 | 0.99 | 1.00 | 0.96 |
| LATANOPROST | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CIPROFLOXACIN | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| HYDROCORTISONE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ASPIRIN | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| DIAZEPAM | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| PREDNISOLONE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| IOHEXOL | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

Performance Score

0.00 0.25 0.50 0.75 1.00


Metric Distribution Across Methods for gene
Violin = distribution, point = median, bar = IQR (25-75%)
ACCURACY
F1
KAPPA
PRECISION
RECALL
SPECIFICITY
BC-Final
BC-Curated
gemini
grok
llama
BC-EmbedEq
gpt
BC-FuzzyEq
0.00
0.25
0.50
0.75
1.00
Score


Gene metric comparison
BC-Final
BC-Curated
grok
gemini
BC-EmbedEq
llama
gpt
BC-FuzzyEq
1.00
0.97
0.74
0.70
0.62
0.57
0.31
0.12
0.00
0.25
0.50
0.75
1.00
Median Metric Score
accuracy
precision
kappa
f1
recall
specificity

# Gene Synonyms Extraction: Comparative Performance Matrix

Methods ranked by Median F1

Gene

### BC-FuzzyEq
Rank 8 | Med F1: 0.12 | R: 0.07 | P: 1.00 | K: 0.12

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 0.18 | 0.31 | 0.97 | 1.00 | 0.30 |
| CDK1 | 1.00 | 0.09 | 0.17 | 0.96 | 1.00 | 0.16 |
| AIRE | 1.00 | 0.10 | 0.18 | 0.97 | 1.00 | 0.18 |
| NDUFS1 | 1.00 | 0.09 | 0.17 | 0.96 | 1.00 | 0.16 |
| TP53 | 1.00 | 0.07 | 0.12 | 0.95 | 1.00 | 0.12 |
| CDH1 | 1.00 | 0.05 | 0.09 | 0.92 | 1.00 | 0.08 |
| TUBB | 1.00 | 0.06 | 0.12 | 0.94 | 1.00 | 0.11 |
| GCK | 1.00 | 0.05 | 0.10 | 0.93 | 1.00 | 0.09 |
| KRAS | 1.00 | 0.06 | 0.11 | 0.88 | 1.00 | 0.10 |
| CSNK2B | 1.00 | 0.05 | 0.10 | 0.93 | 1.00 | 0.09 |
| ACTB | 1.00 | 0.07 | 0.12 | 0.95 | 1.00 | 0.12 |
| BRCA1 | 1.00 | 0.11 | 0.19 | 0.94 | 1.00 | 0.18 |
| EGFR | 1.00 | 0.11 | 0.19 | 0.94 | 1.00 | 0.18 |
| SLC2A1 | 1.00 | 0.04 | 0.08 | 0.91 | 1.00 | 0.07 |
| NR5A1 | 1.00 | 0.05 | 0.09 | 0.92 | 1.00 | 0.08 |

### gpt-4o-mini
Rank 7 | Med F1: 0.31 | R: 0.18 | P: 1.00 | K: 0.30

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 0.18 | 0.31 | 0.97 | 1.00 | 0.30 |
| CDK1 | 1.00 | 0.18 | 0.31 | 0.97 | 1.00 | 0.30 |
| AIRE | 1.00 | 0.10 | 0.18 | 0.97 | 1.00 | 0.18 |
| NDUFS1 | 1.00 | 0.09 | 0.17 | 0.96 | 1.00 | 0.16 |
| TP53 | 1.00 | 0.40 | 0.57 | 0.97 | 1.00 | 0.56 |
| CDH1 | 1.00 | 0.23 | 0.37 | 0.94 | 1.00 | 0.35 |
| TUBB | 1.00 | 0.50 | 0.67 | 0.97 | 1.00 | 0.65 |
| GCK | 1.00 | 0.10 | 0.18 | 0.93 | 1.00 | 0.17 |
| KRAS | 1.00 | 0.46 | 0.63 | 0.93 | 1.00 | 0.59 |
| CSNK2B | 1.00 | 0.45 | 0.62 | 0.96 | 1.00 | 0.60 |
| ACTB | 1.00 | 0.20 | 0.33 | 0.96 | 1.00 | 0.32 |
| BRCA1 | 1.00 | 0.16 | 0.27 | 0.94 | 1.00 | 0.26 |
| EGFR | 1.00 | 0.16 | 0.27 | 0.94 | 1.00 | 0.26 |
| SLC2A1 | 1.00 | 0.12 | 0.22 | 0.92 | 1.00 | 0.21 |
| NR5A1 | 1.00 | 0.05 | 0.09 | 0.92 | 1.00 | 0.08 |

### llama-3.3-70b-versatile
Rank 6 | Med F1: 0.57 | R: 0.40 | P: 1.00 | K: 0.55

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 0.64 | 0.78 | 0.99 | 1.00 | 0.77 |
| CDK1 | 1.00 | 0.64 | 0.78 | 0.99 | 1.00 | 0.77 |
| AIRE | 1.00 | 0.70 | 0.82 | 0.99 | 1.00 | 0.82 |
| NDUFS1 | 1.00 | 0.09 | 0.17 | 0.96 | 1.00 | 0.16 |
| TP53 | 1.00 | 0.33 | 0.50 | 0.96 | 1.00 | 0.49 |
| CDH1 | 1.00 | 0.59 | 0.74 | 0.97 | 1.00 | 0.73 |
| TUBB | 1.00 | 0.50 | 0.67 | 0.97 | 1.00 | 0.65 |
| GCK | 1.00 | 0.40 | 0.57 | 0.96 | 1.00 | 0.55 |
| KRAS | 1.00 | 0.54 | 0.70 | 0.94 | 1.00 | 0.67 |
| CSNK2B | 1.00 | 0.30 | 0.46 | 0.95 | 1.00 | 0.44 |
| ACTB | 1.00 | 0.47 | 0.64 | 0.97 | 1.00 | 0.62 |
| BRCA1 | 0.88 | 0.37 | 0.52 | 0.95 | 1.00 | 0.50 |
| EGFR | 1.00 | 0.32 | 0.48 | 0.95 | 1.00 | 0.46 |
| SLC2A1 | 1.00 | 0.25 | 0.40 | 0.93 | 1.00 | 0.38 |
| NR5A1 | 1.00 | 0.27 | 0.43 | 0.94 | 1.00 | 0.41 |

### BC-EmbedEq
Rank 5 | Med F1: 0.62 | R: 0.47 | P: 1.00 | K: 0.60

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CDK1 | 1.00 | 0.73 | 0.84 | 0.99 | 1.00 | 0.84 |
| AIRE | 1.00 | 0.40 | 0.57 | 0.98 | 1.00 | 0.56 |
| NDUFS1 | 1.00 | 0.64 | 0.78 | 0.99 | 1.00 | 0.77 |
| TP53 | 1.00 | 0.67 | 0.80 | 0.98 | 1.00 | 0.79 |
| CDH1 | 1.00 | 0.45 | 0.62 | 0.96 | 1.00 | 0.60 |
| TUBB | 0.83 | 0.62 | 0.71 | 0.97 | 0.99 | 0.70 |
| GCK | 1.00 | 0.30 | 0.46 | 0.95 | 1.00 | 0.44 |
| KRAS | 1.00 | 0.66 | 0.79 | 0.96 | 1.00 | 0.77 |
| CSNK2B | 1.00 | 0.70 | 0.82 | 0.98 | 1.00 | 0.81 |
| ACTB | 0.70 | 0.47 | 0.56 | 0.96 | 0.99 | 0.54 |
| BRCA1 | 1.00 | 0.37 | 0.54 | 0.96 | 1.00 | 0.52 |
| EGFR | 1.00 | 0.37 | 0.54 | 0.96 | 1.00 | 0.52 |
| SLC2A1 | 1.00 | 0.38 | 0.55 | 0.94 | 1.00 | 0.52 |
| NR5A1 | 0.67 | 0.36 | 0.47 | 0.93 | 0.98 | 0.44 |

### gemini-2.5-flash-lite
Rank 4 | Med F1: 0.70 | R: 0.53 | P: 1.00 | K: 0.68

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 0.90 | 0.82 | 0.86 | 0.99 | 1.00 | 0.85 |
| CDK1 | 1.00 | 0.73 | 0.84 | 0.99 | 1.00 | 0.84 |
| AIRE | 1.00 | 0.60 | 0.75 | 0.99 | 1.00 | 0.74 |
| NDUFS1 | 1.00 | 0.64 | 0.78 | 0.99 | 1.00 | 0.77 |
| TP53 | 1.00 | 0.53 | 0.70 | 0.97 | 1.00 | 0.68 |
| CDH1 | 1.00 | 0.73 | 0.84 | 0.98 | 1.00 | 0.83 |
| TUBB | 1.00 | 0.50 | 0.67 | 0.97 | 1.00 | 0.65 |
| GCK | 1.00 | 0.65 | 0.79 | 0.97 | 1.00 | 0.77 |
| KRAS | 1.00 | 0.40 | 0.57 | 0.92 | 1.00 | 0.54 |
| CSNK2B | 1.00 | 0.55 | 0.71 | 0.97 | 1.00 | 0.69 |
| ACTB | 0.64 | 0.47 | 0.54 | 0.96 | 0.98 | 0.52 |
| BRCA1 | 1.00 | 0.26 | 0.42 | 0.95 | 1.00 | 0.40 |
| EGFR | 1.00 | 0.32 | 0.48 | 0.95 | 1.00 | 0.46 |
| SLC2A1 | 0.06 | 0.38 | 0.10 | 0.39 | 0.39 | -0.07 |
| NR5A1 | 0.00 | 0.00 | 0.00 | 0.92 | 1.00 | 0.00 |

### grok-4.1-fast-non-reasoning
Rank 3 | Med F1: 0.74 | R: 0.59 | P: 1.00 | K: 0.73

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 0.82 | 0.90 | 0.99 | 1.00 | 0.90 |
| CDK1 | 1.00 | 0.82 | 0.90 | 0.99 | 1.00 | 0.90 |
| AIRE | 1.00 | 0.90 | 0.95 | 1.00 | 1.00 | 0.95 |
| NDUFS1 | 1.00 | 0.82 | 0.90 | 0.99 | 1.00 | 0.90 |
| TP53 | 1.00 | 0.67 | 0.80 | 0.98 | 1.00 | 0.79 |
| CDH1 | 1.00 | 0.59 | 0.74 | 0.97 | 1.00 | 0.73 |
| TUBB | 1.00 | 0.62 | 0.77 | 0.98 | 1.00 | 0.76 |
| GCK | 1.00 | 0.70 | 0.82 | 0.98 | 1.00 | 0.81 |
| KRAS | 1.00 | 0.37 | 0.54 | 0.92 | 1.00 | 0.51 |
| CSNK2B | 1.00 | 0.35 | 0.52 | 0.95 | 1.00 | 0.50 |
| ACTB | 1.00 | 0.40 | 0.57 | 0.97 | 1.00 | 0.56 |
| BRCA1 | 0.90 | 0.47 | 0.62 | 0.96 | 1.00 | 0.60 |
| EGFR | 1.00 | 0.42 | 0.59 | 0.96 | 1.00 | 0.57 |
| SLC2A1 | 0.91 | 0.42 | 0.57 | 0.94 | 1.00 | 0.55 |
| NR5A1 | 1.00 | 0.55 | 0.71 | 0.96 | 1.00 | 0.69 |

### BC-Curated
Rank 2 | Med F1: 0.97 | R: 0.95 | P: 1.00 | K: 0.97

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CDK1 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| AIRE | 1.00 | 0.90 | 0.95 | 1.00 | 1.00 | 0.95 |
| NDUFS1 | 1.00 | 0.91 | 0.95 | 1.00 | 1.00 | 0.95 |
| TP53 | 1.00 | 0.93 | 0.97 | 1.00 | 1.00 | 0.96 |
| CDH1 | 1.00 | 0.86 | 0.93 | 0.99 | 1.00 | 0.92 |
| TUBB | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| GCK | 1.00 | 0.95 | 0.97 | 1.00 | 1.00 | 0.97 |
| KRAS | 1.00 | 0.97 | 0.99 | 1.00 | 1.00 | 0.98 |
| CSNK2B | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ACTB | 1.00 | 0.87 | 0.93 | 0.99 | 1.00 | 0.92 |
| BRCA1 | 1.00 | 0.95 | 0.97 | 1.00 | 1.00 | 0.97 |
| EGFR | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| SLC2A1 | 1.00 | 0.92 | 0.96 | 0.99 | 1.00 | 0.95 |
| NR5A1 | 1.00 | 0.95 | 0.98 | 1.00 | 1.00 | 0.97 |

### BC-Final
Rank 1 | Med F1: 1.00 | R: 1.00 | P: 1.00 | K: 1.00

| Gene | PRECISION | RECALL | F1 | ACCURACY | SPECIFICITY | KAPPA |
|---|---|---|---|---|---|---|
| BAX | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CDK1 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| AIRE | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| NDUFS1 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| TP53 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CDH1 | 1.00 | 0.91 | 0.95 | 0.99 | 1.00 | 0.95 |
| TUBB | 0.89 | 1.00 | 0.94 | 0.99 | 0.99 | 0.94 |
| GCK | 1.00 | 0.95 | 0.97 | 1.00 | 1.00 | 0.97 |
| KRAS | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| CSNK2B | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| ACTB | 0.83 | 1.00 | 0.91 | 0.99 | 0.99 | 0.90 |
| BRCA1 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| EGFR | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| SLC2A1 | 1.00 | 0.96 | 0.98 | 1.00 | 1.00 | 0.98 |
| NR5A1 | 0.84 | 0.95 | 0.89 | 0.98 | 0.98 | 0.88 |

Performance Score

0.00 0.25 0.50 0.75 1.00